



Unit 4

Hongmei Pang

Though this be madness, yet there is method in it
– Shakespeare
Hamlet
Act II, scene i



- All about methods
- Visualize the method
- Rewrite
- Collocation



Methods

- Methods
- Methods and Materials
- Method Description and Validation

Functions:

- to **provide the information** needed for another competent scientist to repeat the work
- to **describe reasoning** for specific procedures or techniques utilized to identify and analyze data
- to allow reader to critically determine the **validity** of the research

- Make sure the descriptions are **complete** and yet are also as **concise** as possible
- Write extremely **clearly**

Heading	What to include
Participants	•Participant or subject characteristics, demographic characteristics •Sample size •Sampling procedures
Materials	•Primary and secondary measures •Quality of measurements
Procedure	•Data collection methods •Research design (e.g., experimental, correlational, or descriptive) •Data processing and diagnostics (e.g., outlier removal) •Data analysis strategy (e.g., comparison or regression tests)

Heading	What to include
Participants	•Participant or subject characteristics, demographic characteristics (like age, sex, ethnicity, religion, education level, and socioeconomic status etc.) •Sample size •Sampling procedures ✓ The participants selection method employed ✓ Reasons why the participants took part in the research study ✓ Where the study was advertised ✓ Whether an incentive was given for participation •Example: Reporting sampling procedures Ethics approval was obtained before we began recruiting participants. Current first-generation college students were invited to participate. The study was advertised through general emails sent to university-wide mailing lists, social media posts, and flyers across campus. Participants were self-selected and compensated \$10 for their time in the hour-long study. •Example: Reporting sample size A total of 200 workers were randomly selected from tech companies.

Heading	What to include
Materials	•Primary and secondary measures ✓ technical equipment, testing instruments, and other materials used in conducting the research •Quality of measurements ✓ training researchers to collect data reliably, ✓ using multiple people to assess (e.g., observe or code) the data, ✓ translation and back-translation of research materials, ✓ using pilot studies to test your materials on unrelated samples. • Example: Reporting materials The Academic Anxiety Inventory (AAI; Pizzie & Kraemer, 2017) was used to measure academic anxiety in college students. The inventory consists of 50 Likert scale questions with subscales for math, science, and test anxiety. The participants indicated their level of agreement with statements on a 5-point scale ranging from “strongly disagree” to “strongly agree.” An example item is “I feel anxious when studying for a final.”

Heading	What to include
Procedure	•Data collection methods ✓ surveys, interviews, observations, focus groups, neuroimaging, cognitive tests, bibliometric analysis and so on. •Research design (e.g., experimental, correlational, or descriptive) • Example: Reporting data collection methods and research design All participants were told that the survey concerned students' general knowledge and would take a maximum of thirty minutes. After arriving at the laboratory individually, they were assured confidentiality, and they provided informed consent. Participants were randomly assigned to a control or experimental condition. The survey was administered using Qualtrics (https://www.qualtrics.com). To begin, all participants were given the AAI and a demographics questionnaire to complete, followed by an unrelated filler task. In the control condition, participants completed a short general knowledge test immediately after the filler task. In the experimental condition, participants were asked to visualize themselves taking the test for 3 minutes before they actually did. For more details on the exact instructions and tasks given, see supplementary materials.

Heading	What to include
Procedure	•Data processing and diagnostics (e.g., outlier removal)
	• Example: Data cleaning At the beginning of the research, it was necessary to clean the data to create a relevant basis for further analysis. The cleaning phase was done manually using Microsoft Excel. Firstly, all words that were not written in the English language were translated into it. After that, synonyms that were written in British and American English were brought down to one common term – written in American English. The next challenge that needed to be solved was to eliminate abbreviations of some words. Such text was merged with its full name and deleted from the list of independent terms for analysis. ...
	•Data analysis strategy (e.g., comparison or regression tests)
	• Example: Reporting analytical strategy First, we assessed whether there were any baseline differences between the two groups in terms of demographics characteristics or AAI scores. Subsequently, to test our primary hypothesis that the visualization intervention improves test performance, we performed an independent samples <i>t</i> test on test scores.

Style

- more use of third person, passive constructions than others
- use the **past tense** throughout (your discipline and your chosen journal)
 - The present simple is also often used when you are presenting your procedure, model, software, device etc. In this case the active form is often used.
 - Firstly, we define x as an exogenous measure of the natural rate of longevity of people.
 - Our machine uses diesel ... It has a 1000 hp engine ...
 - The application requires 10 TB of space ...
- **not** a step-by-step, directive, protocol as lab manual

Disciplinary variation in methods

- All RPs have introductory and concluding material, but not all have methodological material (humanistic essays, theory papers, certain analytical papers, definitional papers)
- Disciplines vary in the amount of detail expected (what to include & what **not** to include)
- Disciplines vary in what needs to be included (equipment? statistics? sources? permissions? sponsorship? inclusion criteria?)

Disciplinary variation

Method Section Features:

Clipped <-> Intermediate <-> Elaborated



Features of a Clipped text:



- Avoids named sub-sections
- Uses acronyms and citations as shorthand for procedural descriptions
- Assumes background knowledge of the general methodology
- Uses on occasion a running series of verbs in a sentence (e.g. "...collected, stained and stored")
- Avoids definitions of terms and examples
- Offers few "how" statements, such as "by + verb-ing"
- Provides few or no justifications for methodological choices

The standard **AOAC methods** (AOAC,1975) were used for the determination of total solids, nitrogen, crude fat, ash, and Vitamin C. Total sugars were determined by **the method of Potter et al.** (1968), and the total carbohydrates (in terms of glucose) were assayed according to the **procedure of Dubois et al.** (1956). The **method of Kohler and Patten** (1967) was followed for determining amino acid composition.

Quoted by Knorr-Cetina 1981,157

- Standard practices
- Established methods
- Assume the reader has the basic knowledge of the relevant technique.

Elaborated methods sections:



- Recognize a need to provide background knowledge
- Frequently contain sub-sections
- Use descriptions rather than citations to indicate the various aspects of the methodology adopted
- Provide definitions, examples and illustrations as seems necessary
- Include justifications and rationales for details of the procedures adopted, sometimes placed in the marked pre-subject position via a purpose clause
- Often contain one or more of *by+V-ing* "how statements"
- Tend to have a wider range of linking phrases (logical, temporal and spatial) at the beginning of sentences

Effects of Social Media Usage and Social Media Multitasking on the Academic Performance of University Students

Procedure

Mass emails were sent to all undergraduate students at a comprehensive university in Hong Kong to invite them to participate on a voluntary basis (Jupp, 2006). While this nonprobability sampling method may not guarantee the representativeness of the sample collected, it is the most convenient approach to reaching a wide variety of participants from all faculties of the university. To attract more participants, a lucky draw to win HKD100 cash coupons was offered. Anonymous data were collected from the participants through an online survey website. The participants first completed questionnaires measuring their social media usage, SMM, and academic performance. They then provided some demographic information such as gender, age, faculty, degree major, year of study, computer or Internet experience, and information technology (IT) proficiency.

Participants

The participants were 348 undergraduate students from eight university faculties. Informed consent was obtained from all participants prior to data collection. There were 109 males and 232 females in the sample and their ages ranged from 17 to 28 years (mean = 20.252, SD = 1.565). They studied in the faculties of arts (N = 51), business administration (N = 73), education (N = 19), engineering (N = 29), law (N = 8), medicine (N = 53), science (N = 47), and social science (N = 62). There were 123 first-year, 85 second-year, 75 third-year, 57 fourth-year, and 2 fifth-year students. They had, on average, 12–14 years of experience in using computers or the Internet and regarded their IT proficiency to be good. Between 1.4% and 2.0% of the data were missing for the aforementioned demographic variables.

Measures

The Social Media Learning Scale developed and refined by Mills, Knezek, and Wakefield (2013) was used to assess student perceptions on the application of social media to support university learning (SMUL). In particular, the scale measures university student perceptions of using social media for online community learning and building, and is a seven-item unidimensional scale with alpha reliability of 0.74. The items were rated on a 5-point Likert-type scale from 1 (strongly disagree) to 5 (strongly agree).

Academic performance was evaluated with a single item. The participants were required to provide their cumulative GPAs (CGPAs) in an open response format (Paul, Baker, & Cochran, 2012).

Use
subheadings to
organize

Reasons for
using a
particular
methodology

Specify the
purpose and
details of
methods/actions

Sample size;
Parameters/
variables

Standard
methods with
brief
descriptions

Citation for
justification

Disciplinary variation

Clipped

chemistry

materials sc.

mycology

molecular

biology

Intermediate

public health

political science

systematic botany

medical research

applied linguistics

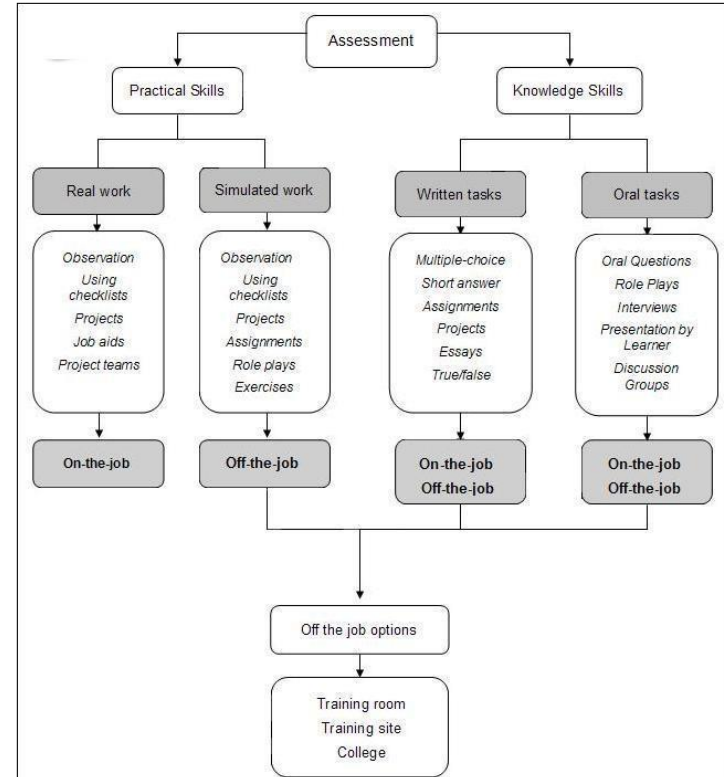
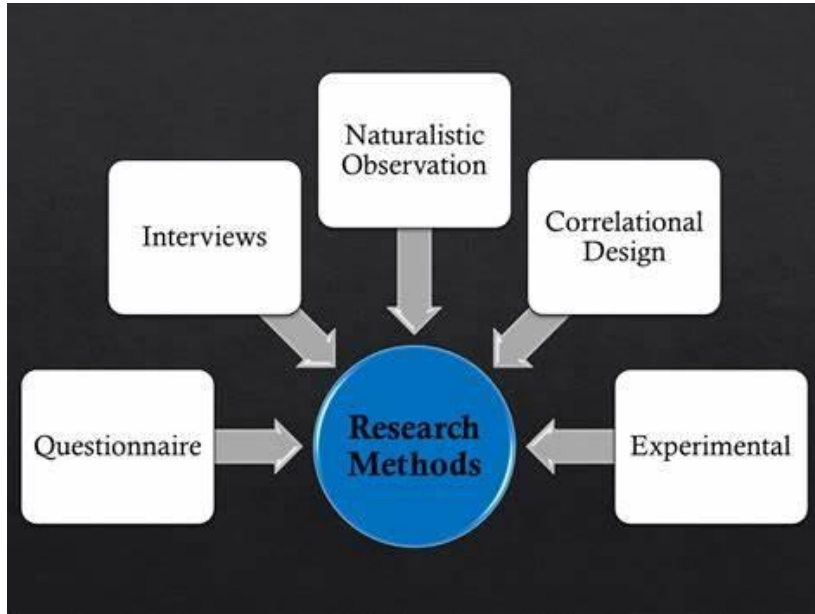
Elaborated

psychology

sociology

education

Visualize your method section



How Do I Love Thee? Let Me Count the Words: The Social Effects of Expressive Writing Methods

Participants

Eighty-six undergraduate psychology students (55 women, 31 men; mean age = 18.7, $SD = 1.0$) and their partners (31 women, 55 men; mean age = 19.3, $SD = 1.6$) were recruited on the basis that they were in a committed heterosexual romantic relationship and that they engaged in IM conversation with each other every day. The couples had been dating an average of 1.31 years ($SD = 1.06$).

Procedure

One member of each couple participated in the writing phase of the study. An experimenter met with participants individually or in small groups of 2 to 5 to give them an overview of the study. They were told that its purpose was to quantify the types of words that people use in writing and everyday interactions.

Participants were instructed to forward to a secure e-mail address all daily IM conversations between themselves and their partners during the 10 days of the study. Considerable effort was taken during the introductory session to reassure participants that their messages would be completely confidential.

Questionnaires assessing demographic information, relationship status, and relationship satisfaction were completed by participants and their partners on-line from home on the first day of the study. Satisfaction was measured using the Relationship Assessment Scale (RAS; Hendrick, 1988). The RAS consists of seven items, such as "In general, how satisfied are you with your relationship?" Ratings are made on a 7-point Likert scale. Participants were randomly assigned to the experimental ($n = 44$) or control ($n = 42$) writing condition and were instructed to set aside 20 min per day on the fourth, fifth, and sixth days of the study for completion of the on-line writing exercises. Participants in the experimental condition were instructed to write about their deepest thoughts and feelings about their current romantic relationship, "to really let go and explore your very deepest emotions and thoughts about your relationship." Those in the control condition were instructed to write in detail about their daily activities—a standard control condition for expressive-writing studies. The purpose of the writing assignments was left intentionally vague to reduce the possibility of demand characteristics; participants were told that the writing assignments would be used to assess basic word use. Three months later, participants completed a brief on-line questionnaire that assessed relationship status and satisfaction.

Linguistic Analysis

The IM conversations were analyzed using the Linguistic Inquiry and Word Count program (LIWC; Pennebaker, Francis, & Booth, 2001). All conversations from couples in both conditions were spell-checked prior to being submitted to LIWC. They were converted to text files, categorized by speaker (participant or partner), and subcategorized according to whether they dated from before or after the writing manipulation, resulting in a total of four separate IM text files per couple.

How Do I Love Thee?

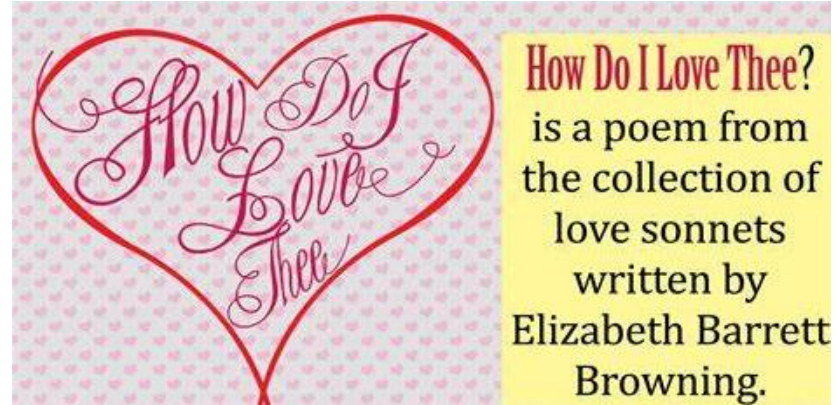
---by Elizabeth Barrett Browning

How do I love thee? Let me count the ways.
I love thee to the depth and breadth and height
My soul can reach, when feeling out of sight
For the ends of Being and ideal Grace.

I love thee to the level of everyday's
Most quiet need, by sun and candlelight.
I love thee freely, as men strive for Right;
I love thee purely, as they turn from Praise.

I love thee with the passion put to use
In my old griefs, and with my childhood's faith.

I love thee with a love I seemed to lose
With my lost saints,--I love thee with the breath,
Smiles, tears, of all my life!--and, if God choose,
I shall but love thee better after death.



Participants

Procedure

Measures

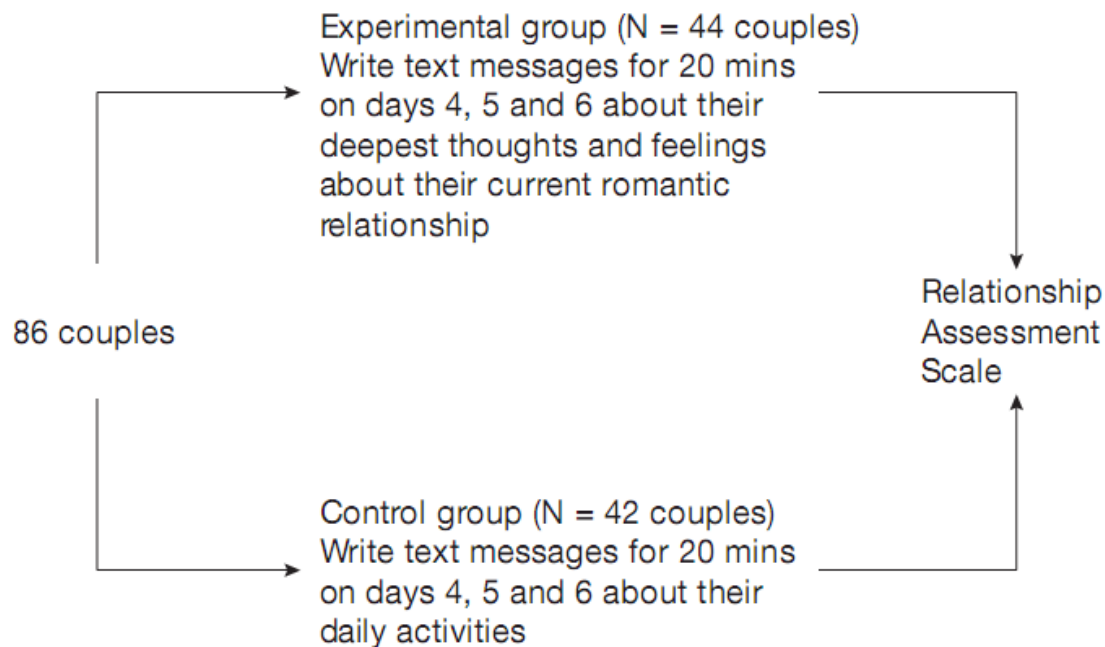


Figure 2.6.1 A schematic illustration of the prose version of the Method used in the study by Slatcher and Pennebaker (2006).

[illegible]

Elizabeth Everett Browning

Abstract

Writing about emotional experiences is associated with a host of positive outcomes. This study extended the expressive-writing paradigm to the realm of romantic relationships to examine the social effects of writing. For 3 consecutive days, one person from each of 86 dating couples either wrote about his or her deepest thoughts and feelings about the relationship or wrote about his or her daily activities. In the days before and after writing, instant messages were collected from the couples. Participants who wrote about their relationship were significantly more likely to still be dating their romantic partners 3 months later. Linguistic analyses of the instant messages revealed that participants and their partners used significantly more positive and negative emotion words in the days following the expressive-writing manipulation if the participants had written about their relationship than if they had written about their daily activities. Increases in positive emotion words partially mediated the relation between expressive writing and relationship stability.

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I have been collecting sales receipts for several years now, ever since receiving a computerized receipt from a taxi-driver in Stockholm in 2000, which first aroused my interest in these small pieces of paper. My collection now totals around 100 examples from 15 countries. I have analysed them in terms of the extra elements they contain, such as web-sites, names of servers, expressions of gratitude, and exhortations to return. I have used examples in workshops in various countries in order to get various kinds of feedback. I also believe that these receipts make excellent ESP teaching materials at several proficiency levels.



How to make it more academic?



During the 2000-2010 period, 103 sales receipts were obtained from vendors in 15 countries, including Chile in South America, and Croatia and Sweden in Europe. A coding system was developed in order to tabulate elements such as expressions of gratitude, calculations of tax, web references, and precise time of sale. In addition, a standard move analysis was conducted, designed to show the preferred orders of placement of such elements. Finally, focus group information was obtained, allowing the researcher to trace similarities and differences in national responses to this evolving genre.



My character
in the first
draft.



My character
after 20
rewrites.

WWW.WRITEREDWIRE.CO.ZA



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1. ~~First person~~
2. ~~Voice~~
3. ~~Tense~~
4. **Sample: Precise number; time frame; sources**
5. **Approach and procedure**
6. **Connecting words**
7. ~~Irrelevant information~~

8. Vocabulary choice

- collect
- website
- workshop
- various kinds of feedback
- small pieces of paper
- I

- obtain
- web reference
- focus group
- similarities and differences in national responses
- evolving genre
- the researcher

六、 研究方法

1、本论文采用通宵作战法，以

Reading research papers



Introduction

Differences Between Potential and Reality

1. the well-established potential of digital technologies
2. survey population: the “digital native”
3. inconsistency: realistic sense (rationale)
4. Research Questions
 - What forms of digital technology use do undergraduate students report as being notably helpful or beneficial in terms of their university studies?
 - What are the characteristics and contexts of students' beneficial technology use?
 - What meanings and wider connotations related to university study and the student experience are associated with these perceived benefits?

the specific practices

factors that influence
how and when
students use digital
technologies

implications of
students' perceptions

Methods

- Participants:
 - sample size and source: a total of 1658 undergraduate students from two universities
 - representativeness: various academic years, disciplines, modes of study, age groups, and academic performance levels
 - Biases: an over-representation of female students and students from University A. However, the analysis indicated that these biases did not significantly affect the results.
- Research Methods:
 - open-ended questions: what +why (nominate and justify)
 - 4594 different examples, resulting in a data set totaling 103,299 words.
 - thematic analysis of the data: 11 distinct digital practices and coding them accordingly

Methods : Theoretical saturation

- It refers to obtaining enough codes to reach the point where they are sufficient to answer research questions.

How Many Interviews Are Enough?: An Experiment with Data Saturation and Variability

Guidelines for determining nonprobabilistic sample sizes are virtually nonexistent. Purposive samples are the most commonly used form of nonprobabilistic sampling, and their size typically relies on the concept of “saturation,” or the point at which no new information or themes are observed in the data. Although the idea of saturation is helpful at the conceptual level, it provides little practical guidance for estimating sample sizes, prior to data collection, necessary for conducting quality research. Using data from a study involving sixty in-depth interviews with women in two West African countries, the authors systematically document the degree of data saturation and variability over the course of thematic analysis. They operationalize saturation and make evidence-based recommendations regarding nonprobabilistic sample sizes for interviews. Based on the data set, they found that **saturation occurred within the first twelve interviews, although basic elements for metathemes were present as early as six interviews.** Variability within the data followed similar patterns.

Results

1. logistics of learning management
2. flexibility of place and location
3. time-saving
4. reviewing, replaying, and revising
5. information research
6. supporting basic tasks
7. communication and collaboration
8. augmenting university learning materials
9. seeing information in different ways
10. cost-saving
11. assessing progress

Discussion

- ‘best practice’ and ‘most useful’ → bounded nature of university students’ engagements with digital technology (para.23)
- **The role of digital technologies:**
 - logistics of study vs learning per se (para.24)
 - strategic use of technology(para.26) :
 - outcome-led practices like completing assignments and getting good grades, “surface” learning
 - not engage with knowledge, not enhance learning
- **Constraints of current university contexts:**
 - curricula, content, teaching and evaluation methods (para.28)

Conclusions

- **Findings:** disconnect between rhetoric and reality
 - acknowledgment of digital technology's role
 - limited transformative impact
 - “safe, bounded and outcome-focused use”
 - “not the most expansive, expressive, empowering, enlightening or even exciting ways”
- **Implications**
 - support for existing uses
 - call for broader institutional changes: alternative learning environments that support “more active, participatory, and creative” uses of digital technology



Work on the collocations.

- ① 不可缺少的部分
- ② 一个反复出现的主题
- ③ 临时小组活动
- ④ 因此
- ⑤ 数字原住民

- ① an integral aspect
- ② a recurring theme
- ③ ad hoc group activities
- ④ as such
- ⑤ digital native

Learning achievements

- ✓ All about methods
- ✓ Visualize the method
- ✓ Rewrite
- ✓ Collocation

